

An Elementary Obstruction to Uniform Prime-Gap Asymptotics and the Additive Structure of the Admissible Constants

Draft note

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Abstract

Let p_n be the n -th prime, let $g_n = p_{n+1} - p_n$, and put

$$d(x) = \max_{p_n < x} g_n.$$

For a fixed constant C , consider the uniform asymptotic

$$\pi(y + Cd(x)) - \pi(y) \sim \frac{Cd(x)}{\log y} \quad (x/2 < y < x).$$

We record the elementary obstruction coming from record prime gaps, prove that no $0 < C \leq 1$ can satisfy the asymptotic, and then use the difference observation that constants satisfying the asymptotic are closed under subtraction. It follows that the set of admissible constants is either trivial or a cyclic lattice $C_0\mathbb{Z}$, generated by a unique constant $C_0 > 1$.

1 The uniform assertion

Let p_n denote the n -th prime and write

$$g_n = p_{n+1} - p_n$$

for the corresponding prime gap. For $x > 2$, define

$$d(x) = \max_{p_n < x} (p_{n+1} - p_n).$$

Thus a gap is counted as soon as its left endpoint is below x ; the right endpoint of that gap need not be below x .

For $C \neq 0$, let $A(C)$ denote the assertion

$$\sup_{x/2 < y < x} \left| \frac{\pi(y + Cd(x)) - \pi(y)}{Cd(x)/\log y} - 1 \right| \rightarrow 0 \quad (x \rightarrow \infty). \quad (1)$$

For the structural statement, we adjoin 0 as a formal trivial constant and set

$$\mathcal{C} = \{0\} \cup \{C \neq 0 : A(C) \text{ holds}\}.$$

The ratio in (1) is not meant to define $A(0)$; 0 is included only as the identity element for the additive-structure statement.

We use two standard facts. First, prime gaps are unbounded, hence strict record gaps occur infinitely often. Second, by the prime number theorem,

$$d(x) = o(x). \quad (2)$$

Indeed $p_{n+1}/p_n \rightarrow 1$, so every prime gap with left endpoint below x is $o(x)$.